

SAM-I2V++: Efficiently Upgrading SAM for Promptable Video Segmentation

Haiyang Mei, Pengyu Zhang, Mike Zheng Shou

Abstract—Foundation models like the Segment Anything Model (SAM) have significantly advanced promptable image segmentation in computer vision. However, extending these capabilities to videos presents substantial challenges, particularly in ensuring precise and temporally consistent mask propagation in dynamic scenes. SAM 2 attempts to address this by training a model on massive image and video data from scratch to learn complex spatiotemporal associations, resulting in huge training costs that hinder research and practical deployment. In this paper, we introduce SAM-I2V++, a training-efficient image-to-video upgradation method for cultivating a promptable video segmentation (PVS) model. Our approach strategically upgrades the pre-trained SAM to support PVS, significantly reducing training complexity and resource requirements. To achieve this, we introduce three key innovations: (i) an image-to-video feature extraction upgrader built upon SAM's static image encoder to enable spatiotemporal video perception, (ii) a memory selective associator that retrieves the most relevant past frames via similarity-driven selection and uses multiscale-enhanced cross-attention to associate selected memory features with the current frame, and (iii) a memory-as-prompt mechanism leveraging object memory to ensure temporally consistent mask propagation in dynamic scenes. Comprehensive experiments demonstrate that our method achieves 93% of SAM 2's performance while using only 0.2% of its training cost. Our work presents a resource-efficient pathway to PVS, lowering barriers for further research in PVS model design and enabling broader applications and advancements in the field. Project page: <https://github.com/showlab/SAM-I2V>.

Index Terms—Segment anything, promptable video segmentation, efficient training, temporal feature integration, memory selective association, memory prompt generation.

I. INTRODUCTION

Foundation models have revolutionized artificial intelligence by leveraging massive datasets and large-scale models to achieve remarkable performance across various tasks. In computer vision, the Segment Anything Model (SAM) [1] has emerged as a prominent foundation model for general-purpose promptable image segmentation, enabling users to segment objects/regions in images based on prompts such as points, boxes, or masks. While SAM demonstrates impressive capabilities on static images, extending these abilities to video data introduces significant challenges due to the temporal dynamics and consistency required in video segmentation tasks.

Recently, SAM 2 [5] has been developed to address promptable video segmentation (PVS), aiming to extend SAM's image segmentation capabilities to videos. SAM 2 is trained from

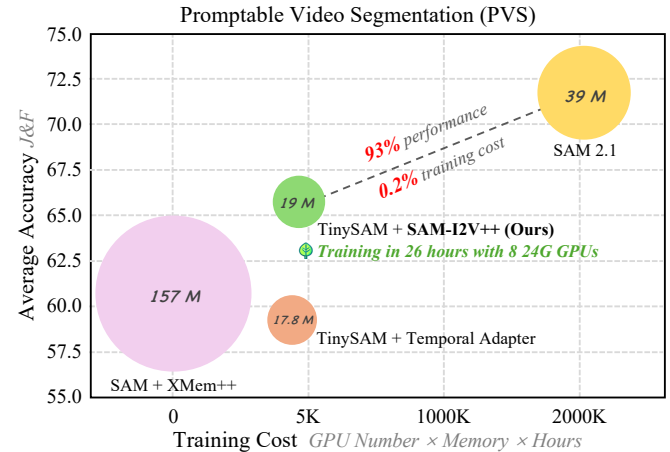


Fig. 1. Comparison on promptable video segmentation (PVS). We depict the online-PVS performance (in J&F) averaged on four benchmark datasets [2]–[5] (vertical axis) with respect to the model training cost (horizontal axis). The size of circles indicates model parameters (in M). SAM 2.1 [5] is a strong PVS model but needs huge training cost (i.e., 256 80GB A100 GPUs for 108 hours). We propose to cultivate a PVS model via image-to-video model upgradation. Building on the pre-trained SAM [1], [6], incorporating existing video object segmentation models (e.g., XMem++ [3]) or temporal association modules (e.g., temporal adapter [5]) significantly reduces training costs compared to SAM 2.1 [5]. Furthermore, our SAM-I2V++ helps achieve 93% of SAM 2.1's performance while using only 0.2% of its training cost.

scratch on a massive collection of images and videos to learn complex spatiotemporal associations. However, this approach demands exorbitant computational resources, requiring 256 80GB A100 GPUs over 108 hours for training. Such high computational costs create substantial barriers to exploring alternative architectures and advancing research in PVS, limiting accessibility for many researchers and practitioners.

This raises a critical question: *Can we achieve promptable video segmentation through a resource-efficient approach?* If so, it would open opportunities for developing diverse PVS foundation models, benefiting a wide range of downstream applications. Addressing this challenge, we propose **SAM-I2V++**, an efficient method to upgrade the pre-trained image-based SAM [1] model to support promptable video segmentation with affordable training cost (Figure 1). We strategically leverage SAM's strong promptable image segmentation capabilities and develop image-to-video upgradation techniques to extend functionality from images to videos.

To bridge the gap between promptable image and video segmentation, our SAM-I2V++ includes three key components: a temporal feature integrator TFI, a memory selective associator MSA, and a memory prompt generator MPG. Specifically, (i) the TFI enhances SAM's image encoder by integrating

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temporal information, enabling the extraction of spatiotemporal features essential for modeling dynamic scenes; (ii) the MSA retrieves most relevant segmentation information from preceding video frames through a new memory filtering mechanism, associating the selected historical information via a multiscale-enhanced cross-attention to guide the current frame's segmentation; and (iii) the MPG incorporates memory as an additional prompt, utilizing object memory to ensure precise and temporally consistent mask propagation across frames. Comprehensive experiments validate that our method significantly reduces training complexity and resource requirements, achieving 93% of SAM 2's performance with only 0.2% of its training cost. In summary, our contributions are:

- 1) We present SAM-I2V++, the first attempt to upgrade a pre-trained image-based SAM model to support promptable video segmentation, enabling PVS capabilities with substantially reduced training costs.
- 2) We design an upgrade method that extends SAM's static image encoder into a video perception module, enabling spatiotemporal feature extraction for PVS.
- 3) We propose a selective memory association strategy that first selects the most relevant past frames via similarity-driven filtering and then associates them with the current frame via multiscale-enhanced cross-attention, facilitating more effective utilization and integration of historical information for memory-guided video segmentation.
- 4) We introduce a memory-as-prompt mechanism that leverages object memory to benefit temporally consistent mask propagation in dynamic scenes.

A preliminary version of this work was presented at CVPR 2025 [7]. This submission extends it in three aspects:

- 1) Upgraded methodology. To enhance promptable video segmentation, we introduce a stronger memory associator by (i) embedding a new scale adapter within cross-attention to handle large and dynamic object-scale variations through multi-scale spatial cues mining and integration, and (ii) streamlining the self-attention block while adding auxiliary mask supervision to efficiently yet effectively regularize temporal memory integration. These innovations collectively establish the advanced SAM-I2V++ framework.
- 2) Improved performance. SAM-I2V++ demonstrates clear advantages over the original SAM-I2V baseline, achieving $J&F$ gains of 3.7, 4.8, and 4.7 points on the challenging SAV-Test set under online-PVS, offline-PVS, and semi-VOS settings, respectively.
- 3) Comprehensive experiments and analysis. We provide a more detailed description for method and experimental setup, supplement extensive scalability studies across various SAM backbones and training budgets, and deepen our analysis and insights with more ablation experiments, discussions of limitations, and future research directions.

II. RELATED WORKS

In this section, we first review promptable visual segmentation for both images and videos, followed by interactive video object segmentation methods that extend user-guided mask

refinement across frames. We then introduce semi-supervised video object segmentation approaches driven by an initial mask, and finally discuss image-to-video transfer learning strategies that adapt pretrained image models for video tasks.

A. Promptable Visual Segmentation

The Segment Anything Model (SAM) [1] has set a significant milestone in promptable segmentation. SAM is designed to generate a valid segmentation mask for an object of interest in an image when provided with input prompts such as bounding boxes or reference points. Trained on the extensive SA-1B dataset, SAM facilitates zero-shot segmentation with its versatile prompting mechanism, making it highly adaptable for diverse downstream tasks. This foundational work has spurred further innovations aimed at enhancing SAM's quality and efficiency. For instance, models like HQ-SAM [8] have focused on improving output precision through high-quality token integration and fine-tuning with detailed masks. Efficiency-oriented models such as EfficientSAM [9], MobileSAM [10], [11], FastSAM [12], TinySAM [6], and SlimSAM [13] have broadened the scope of SAM's applications, enabling its deployment in more resource-constrained and real-world scenarios, including medical imaging [14]–[16] and remote sensing [17]–[19].

Building upon the success of SAM [1], SAM 2 [5] represents an evolution toward promptable video segmentation, addressing the complexities of propagating segmentation across multiple frames. This allows for interaction at any frame of a video, accepting prompts such as positive/negative clicks, bounding boxes, or masks to define or refine object segmentation. Unlike its predecessor, SAM 2 not only responds to initial prompts on single frames but also propagates these segmentation cues throughout the entire video, maintaining temporal consistency and adaptability across frames. Evaluated through interactive online and offline settings, SAM 2's integration of promptable video interaction ensures comprehensive and dynamic segmentation, pushing the boundaries of practical real-world applicability [20]–[30].

B. Interactive Video Object Segmentation

Interactive video object segmentation (IVOS) is an essential task for enabling real-time object tracking and segmentation in videos through user input, such as clicks or bounding boxes. Recent studies [31]–[33] have aimed to develop modular approaches that allow segmenting an object in a single frame and then propagating this segmentation across subsequent frames to maintain consistency. Building upon this, many IVOS methods leverage interactive mechanisms to guide segmentation over multiple frames. While SAM has been used for video object segmentation combined with video trackers based on masks [34]–[36] or points [37], these approaches face certain limitations: trackers may not be effective for all objects, SAM may perform suboptimally on certain video frames, and there is no interactive mechanism for correcting model errors aside from re-annotating erroneous frames and restarting the tracking process from that point. This highlights the expanded functionality of SAM 2 that provides a more flexible solution to enable interactive refinement.